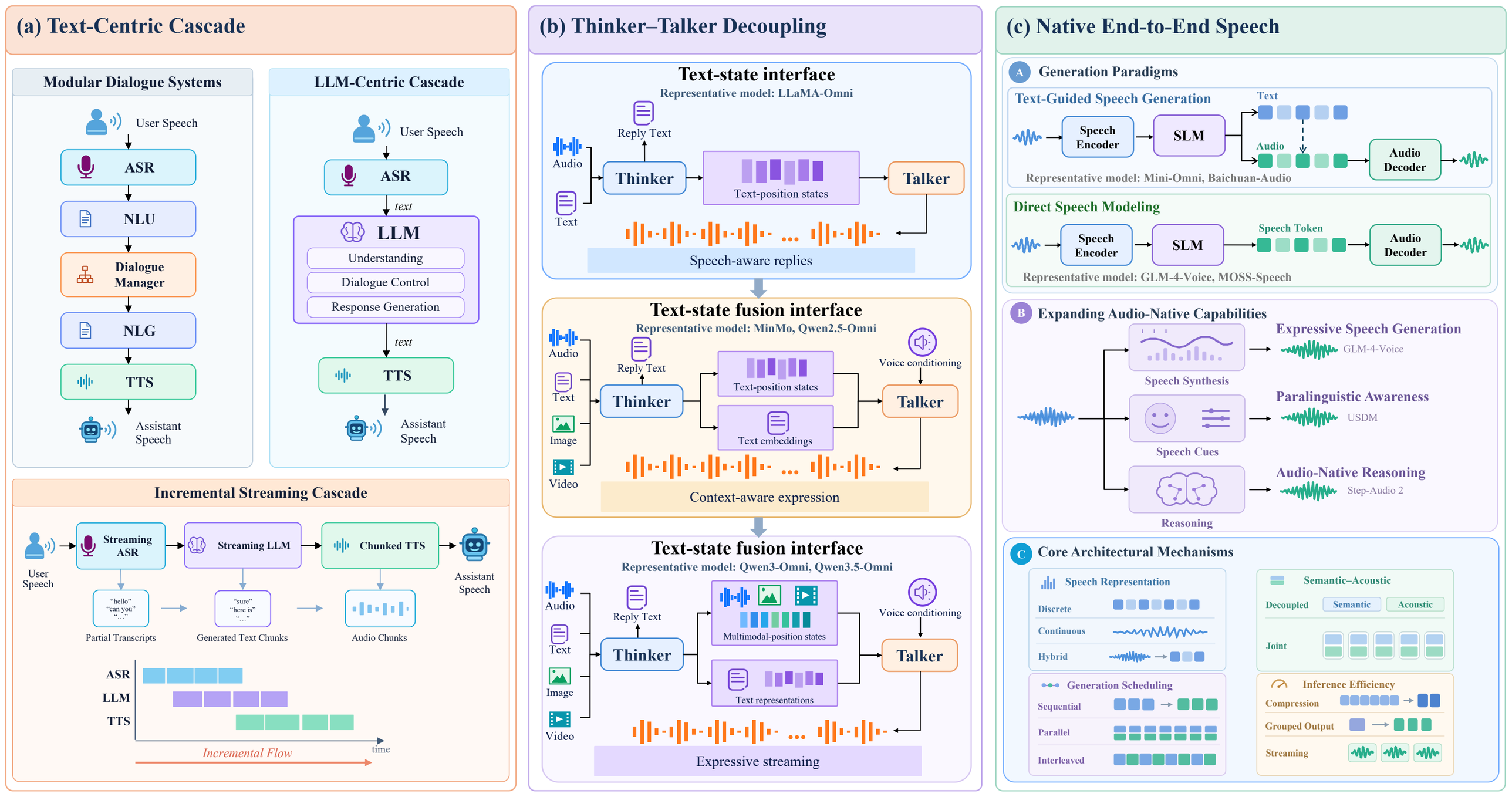

(a) Text-Centric Cascade
Modular Dialogue Systems
User Speech
ASR
NLU
Dialogue Manager
NLG
TTS
Assistant Speech
LLM-Centric Cascade
User Speech
ASR
text
LLM
Understanding
Dialogue Control
Response Generation
text
TTS
Assistant Speech
Incremental Streaming Cascade
User Speech
Streaming ASR
Streaming LLM
Chunked TTS
Assistant Speech
"hello" "can you" "…"
"sure" "here is" "…"
Partial Transcripts
Generated Text Chunks
Audio Chunks
ASR
LLM
TTS
time
Incremental Flow
(b) Thinker–Talker Decoupling
Text-state interface
Representative model: LLaMA-Omni
Reply Text
Audio
Text
Thinker
Text-position states
Talker
Speech-aware replies
Text-state fusion interface
Representative model: MinMo, Qwen2.5-Omni
Audio
Reply Text
Text
Image
Video
Thinker
Text-position states
Text embeddings
Voice conditioning
Talker
Context-aware expression
Text-state fusion interface
Representative model: Qwen3-Omni, Qwen3.5-Omni
Audio
Reply Text
Text
Image
Video
Thinker
Multimodal-position states
Text representations
Voice conditioning
Talker
Expressive streaming
(c) Native End-to-End Speech
A Generation Paradigms
Text-Guided Speech Generation
Speech Encoder
SLM
Text
Audio
Audio Decoder
Representative model: Mini-Omni, Baichuan-Audio
Direct Speech Modeling
Speech Encoder
SLM
Speech Token
Audio Decoder
Representative model: GLM-4-Voice, MOSS-Speech
B Expanding Audio-Native Capabilities
Speech Synthesis
Expressive Speech Generation
GLM-4-Voice
Speech Cues
Paralinguistic Awareness
USDM
Reasoning
Audio-Native Reasoning
Step-Audio 2
C Core Architectural Mechanisms
Speech Representation
Discrete
Continuous
Hybrid
Semantic–Acoustic
Decoupled
Semantic
Acoustic
Joint
Generation Scheduling
Sequential
Parallel
Interleaved
Inference Efficiency
Compression
Grouped Output
Streaming